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Original Article

wm&r

Waste Management & Research
31(8) 783–791
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DOI: 10.1177/0734242X13493959
wmr.sagepub.com
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Aerobic biological pretreatment of municipal solid waste with a high content of putrescibles: effect on landfill emissions

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Abstract

The objective of this work was to study the effect of aerobic biological pretreatment on the emissions of municipal solid waste (MSW) with a high content of putrescibles after landfilling. For this purpose, the organic fraction of MSW was simulated by a mixture of food waste and office paper at a 2.4:1 wet weight ratio. MSW was first pretreated aerobically for three different time periods (8, 45 and 90 days) resulting in organic matter reductions equal to 15%, 45% and 81% respectively. MSW were then placed in 160-L air-tight anaerobic bioreactors. The control anaerobic bioreactors contained untreated MSW. Anaerobic experiments lasted from 300 to 550 days. Results showed that the biogas production from untreated MSW was 372 NL dry kg^{-1} (average of two replications) after 530 days. The MSW that was pretreated aerobically for 45 days and 90 days yielded 130 and 62 NL dry kg^{-1} of biogas after 310 days and 230 days respectively. However, the 8-day (very short-term) pretreatment period led to an increase of the biogas yield (550 NL dry kg^{-1} after 340 days) compared with that of raw refuse. All three runs with aerobically pretreated MSW reached the steady methanogenic phase faster than raw MSW. Leachate emissions were significantly lower in the aerobically-pretreated MSWthan the untreated ones. The leachate ammonium concentrations had an increasing trend in all anaerobic reactors and reached a plateau of between 2 and 3.5 g L^{-1} at the end of the process.

Keywords

Biogas yields, bioreactor landfill, sustainable landfilling, zero emission landfills

Introduction

According to European Directive 2008/98/EC (European Council, 2008), landfilling of municipal solid waste (MSW) is the least desirable management technology. In addition, European Directive 1999/31/EC (European Council, 1999) requires that organics should be diverted from the MSW stream sent to the landfills. In some European countries (e.g. Germany, Sweden), MSW should not be landfilled unless it is extensively pretreated to reduce its organic load below a certain threshold value. For example, in Germany and Sweden, the total organic carbon of MSW prior to landfilling should be less than 18% and 10% (dry weight basis) respectively (Fricke et al., 2005; Praagh et al., 2009). As a result, mechanical–biological pretreatment, and particularly the aerobic biological treatment (ABT), has gained a lot of attention as a pretreatment technique prior to landfilling. The goal of long-term ABT is to produce a stabilized or semi-stabilized product that can be safely disposed of to a zero emission (i.e. final storage or sustainable) landfill (De Gioannis et al., 2009; Scaglia et al., 2010; Schievano et al., 2008; Zach et al., 2000). It is well established that anaerobic emissions of MSW that has been aerobically pretreated differ from those of untreated waste (Brinkmann et al., 1995; Carlsson et al., 2012; Corti et al., 2009; De Gioannis et al., 2009; Güelfo et al., 2011; Komilis et al., 1999; Leikam and Stegmann, 1995; Rieger and Bidlingmaier, 1995; Scaglia et al., 2010; Scharff et al., 1995; Zach et al., 2000). Binner and Zach (1999) performed experiments with laboratory-scale reactors and showed that the 90-day biogas yield of raw MSW decreased from 129 NL dry kg^{-1} to 10 NL dry kg^{-1} after 9 weeks of aerobic biological treatment. Zach et al. (2000) also showed that the 240-day biogas yield decreased by an order of magnitude after 20 weeks of ABT. However, Adani et al. (2004) showed that the 90-day biogas yield (measured in 60-mL vials) of untreated MSW and of MSW treated aerobically for 10 days and 21 days increased from 6 L dry kg^{-1} to 6.8 and 20 L dry kg^{-1} respectively. This finding agrees with previous work by Adani et al. (2000), and can likely be explained by the fact that

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the duration of the anaerobic experiment should have been longer than 90 days to allow the establishment of methanogenesis in the raw refuse samples. Short-term (2-week) aerobic pretreatment has been also found to decrease the methane (CH_4) generation lag time and to accelerate the start up of a dry anaerobic system with the organic fraction of MSW (Brummeler and Koster, 1990); however, the short pretreatment led to a significant loss (40%) of the potential CH_4 yield of the untreated waste.

Despite the amount of research, the effect of aerobic pretreatment on landfill emissions of MSW that contains a high percentage of putrescibles, such as Hellenic MSW (HMD, 2003), has not been investigated. In addition, it appears that there is limited information on the effect of very short-term (e.g. 1 week) aerobic pretreatment periods on the subsequent anaerobic emissions. The objective of this work was, therefore, to quantify gaseous and leachable landfill emissions of aerobically pretreated MSW that contained a high percentage of putrescibles after 8, 45 and 90 days of ABT periods. A relationship between the organic matter (OM) reduction achieved during the ABT phases and the subsequent anaerobic emissions can aid decision makers in determining the fate of the pretreated MSW (i.e. either to obtain a product with enhanced biogas potential for disposal to a bioreactor landfill or a product with reduced anaerobic emissions for disposal to a zero emission landfill).

Materials and methods

Simulation of the organic fraction of MSW

The organic fraction of municipal solid waste (OFMSW) was simulated using a mixture of wasted office paper (OFP) and food waste (FW). OFP was collected from faculty offices and was manually cut to pieces less than 2.5 cm in size. Fifty kilograms of composite FW batches was collected from the university restaurant over a 1-week period. Each batch of FW and OFP was characterized separately prior to the initiation of a run. The simulated OFMSW contained FW and OFP at a 2.4:1 ratio, on a wet weight basis, which is the approximate ratio of these components in the Hellenic MSW (HMD, 2003). We simulated the organic fraction of MSW in order to maintain a consistent and well controlled initial composition of refuse throughout the experiments and to decrease the need of many replicates in our runs. Inorganics were not included in our simulated composition, as they do not contribute to the anaerobic emissions that we measured in this work.

Aerobic pretreatment

The aerobic reactors used to biologically pretreat (compost) the waste were made of high density polyethylene and had a volume of 200 L. A screen was placed 10 cm above the bottom of the reactor to create an air-plenum. Ambient air at a flow rate of 2 L min^{-1} was continuously pumped via a compressor to the bottom of the reactor. Each aerobic digester contained 20 (wet) kg of simulated OFMSW. Thickened aerobic sludge (obtained from a nearby wastewater treatment plant) was added to the mixtures of all aerobic runs to inoculate their contents in order to speed up the process and to increase initial moisture to 55–60% (wet weight basis). The sludge had a moisture content of 98.2% [wet weight (ww)] and an OM content of 80.5% [dry weight (dw)]. The aerobic reactors were filled to two-thirds of their total volume and were placed in a temperature controlled room at 35°C ± 2°C. The aerobic reactor setup is explained in detail in Komilis et al. (2011).

Anaerobic bioreactors

The stainless steel anaerobic bioreactors were air tight and had a volume of 165 L. The OFMSW was placed on top of a screen that was located 6 cm above the bottom of the reactor. The generated biogas exited the bioreactors from a top valve and passed through a cumulative drum-type low flowrate gas meter by Ritter (model TG05/5, Ritter, Bochum, Germany). The gas flowmeters were equipped with a thermometer to measure the temperature of the exiting gas. Actual biogas volumes were converted to normal conditions (0°C, 1 atm) and were reported in NL dry kg^{-1} of mixture. The composition of the gas was measured via a portable gas analyzer (GA94A, Geotechnical Instruments, Leamington, UK) during the same time that the cumulative gas volume was recorded. The gas analyzer was checked periodically using a standard gas of methane (CH_4):carbon dioxide (CO_2) at 49.9%:50.1% (v/v), respectively, and a mixture of oxygen (O_2):nitrogen (N_2) containing O_2 at 8.06% (v/v).

Water was added to the contents of all anaerobic runs to reach an initial moisture content of between 70% and 75% (ww) to stimulate anaerobiosis, to decrease experimental times and to aid in the production of adequate amounts of leachate (Barlaz et al., 1992). Leachate was removed by drainage at the bottom of the reactor every 20–25 days. After collecting small leachate amounts for analysis, the remaining leachate was manually re-circulated back into the reactor from a valve on the top cover. Whenever leachate pH was below 5.5, neutralization was performed by adding aliquots of a 1 N sodium carbonate (Na_2CO_3) liquid solution to reach a close to neutral pH (between 6.8 and 7.2). The anaerobic reactors were designated as R0, R8, R45 and R90, in which the number indicates the duration of the aerobic pretreatment period in days. The total wet masses placed in the anaerobic bioreactors (including the water added to reach the high initial moisture content) ranged from 11.8 kg (R90) to 50.2 kg (R0a), which corresponded to dry masses equal to 3.0 and 12.7 kg respectively. The amounts of the aerobically pretreated end-products that were placed in the subsequent anaerobic reactors varied as a result of the different dry weight reductions achieved within each aerobic reactor. The bulk density of the material placed in the anaerobic reactors ranged from 400 (R0) to 600 kg m^{-3} (R90) and depended on the extent of the aerobic decomposition. The anaerobic experiments were terminated when the biogas production rates were kept below 0.2 NL dry $kg^{-1}d^{-1}$ for five successive days. All anaerobic experiments were performed in a temperature controlled room at the mesophilic temperature range (35°C ± 2°C). The anaerobic reactors were run on a batch mode and simulated a MSW landfill at optimal moisture contents. The reproducibility of the anaerobic

Table 1. Properties of source materials.

	Moisture content (% ww) mean ± SD (n = 3)	Organic matter (% dw) mean ± SD (n = 3)	Carbon (% dw) mean ± SD (n = 5)	Nitrogen (% dw) mean ± SD (n = 5)
FW	65 ± 0.98	92 ± 0.74	48.9 ± 2.6	4.2 ± 0.27
OFP	5.6 ± 0.06	80.9 ± 0.40	34.8 ± 0.38	0.12 ± 0.002
Mixture of FW and OFP (simulated MSW)	50	86	41	2.0

ww: wet weight; dw: dry weight; FW: food waste; OFP: office paper; MSW: municipal solid waste.

Table 2. Properties of the aerobic experiments and calculated solids reduction.

Aerobic pretreatment duration (days)	OM concentration at the end of the aerobic experiment (% dw)[a]	Total carbon concentration at the end of the aerobic experiment (% dw)[a]	Dry matter decrease (%)	OM decrease (%)	Total carbon decrease (%)	Total nitrogen decrease (%)
8	83.6[b] ± 0.20	39.7[b] ± 2.1	13.0	15.2	11.8	10.5
45	78.5[c] ± 0.78	41.0[b] ± 1.8	38.7	44.6	52.2	47.1
90	53.7[d] ± 2.4	28.7[c] ± 2.8	69.1	80.6	80.8	59.0

OM: organic matter; dw: dry weight.
[a]Means ± SD.
[b–d]Means that do not share a letter are statistically different at $P < 0.05$ using Tukey's pairwise comparison test (at n=3 for organic matter and n=5 for carbon and nitrogen); solids reduction was calculated using the ash conservation principle.

system setup was tested by running two replicate anaerobic reactors (R0a, R0b) at different time periods under the same conditions, and with precisely the same source materials and amounts.

Analytical methods

Solids sampling was performed using a sequential quartering procedure. The moisture content was measured at 75°C until constant weight. The dried solids were then ground using a solids' cutting mill (RETSCH, Haan,Germany) with a 1.5-mm mesh size screen and the OM content was measured via the loss on ignition at 550°C for 2 h. Elemental analysis was performed using an elemental analyzer (Thermo Electron, Waltham, MA, USA, model: EA-1110 CHNS-O) according to Komilis et al. (2012). Leachate pH was measured with a portable pH-meter (WTW, Weilheim, Germany). The total 5-day biochemical oxygen demand (BOD_5), total chemical oxygen demand (COD) and ammonium ($N\text{-}NH_4^+$) concentrations of the leachates were measured using standard methods (APHA and AWWA, 1998). The dry matter and OM reduction achieved during the aerobic and anaerobic experiments were calculated based on the principle of ash conservation.

Results and discussion

Degradation of solid waste in the aerobic pretreatment reactors

The properties of the source materials used to simulate the MSW are shown in Table 1. At the end of the aerobic experiments (i.e. after 8, 45 and 90 days), the final moisture contents of the remaining substrates ranged from 45% to 66% (ww). The properties of the end materials and the dry matter reduction calculated during the aerobic experiments are included in Table 2. According to Table 2, the higher the extent of aerobic pretreatment, the higher were the losses of dry mass, OM, carbon and nitrogen within the aerobic reactors. The final OM content of the aerobically pretreated mixtures ranged from 84% dw to 54% dw for the 8- and 90-day aerobic pretreatment period runs. As expected, the 90-day aerobic pretreatment period led to the highest dry matter and OM reduction (69% and 81% respectively) (see Table 2). The total nitrogen reduction achieved during the aerobic phases ranged from 10.5% (8-day pretreatment) to 59% (90-day aerobic pretreatment). Gas content measurements that were performed in the headspace of all aerobic reactors showed that the CO_2 content peaked at 5–6% v/v after 4–5 days and O_2 content reached a minimum of 11–13% v/v at the same time. No CH_4 was detected throughout the aerobic pretreatment process. The good replication of the aerobic reactor setup had been confirmed in a previous study (Komilis et al., 2011).

Biogas yields and composition

Table 3 includes the biogas yields measured at the end of the five anaerobic runs.

Figures 1 and 2 illustrate the normalized CH_4 and CO_2 cumulative yields of all five anaerobic experiments versus time. According to Figure 1 and Table 3, reproducibility of the anaerobic system was relatively weak, as the biogas yields from replicate runs R0a and R0b differed by 40% after 530 days. A difficulty in replicating pilot-scale anaerobic reactors was also noted by Barlaz et al. (1987) during landfill simulation experiments with 208-L drums. Figure 1 reveals that the onset of the steady methanogenesis occurred at 160 days for R0a, whilst it occurred at 280

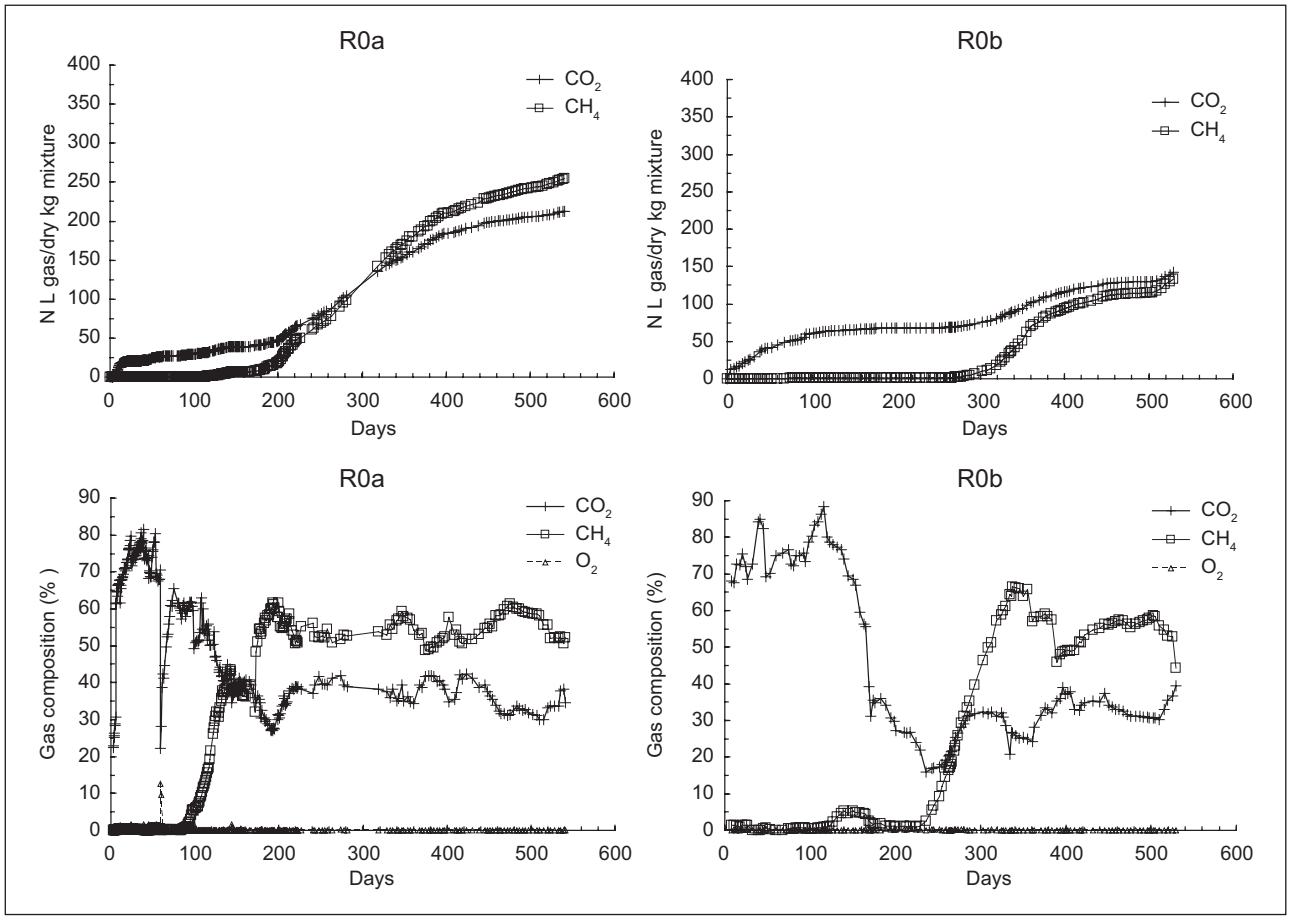


Figure 1. Cumulative methane (CH_4) and carbon dioxide (CO_2) yields (top) and biogas composition (bottom) for (replicate) anaerobic reactors R0a and R0b.

Table 3. Gas yields and solids reduction during the anaerobic experiments.

Run[a]	Duration of anaerobic run (days)	CH_4 yield[b]	CO_2 yield[b]	OM content of the anaerobic residue (% dw)[c]	Carbon content of the anaerobic residue (% dw)[c]	Nitrogen content of the anaerobic residue (% dw)[c]	Dry matter decrease (%)	Organic matter decrease (%)	Carbon decrease (%)	Nitrogen decrease (%)
R0a	541	254	213	67.1[d] ± 1.2	39.3[e] ± 3.1	1.5[g] ± 0.22	58.3	67.6	66.0	72.2
R0b	530	160	171	76.8[e] ± 0.51	40.6[e] ± 2.3	0.75[d] ± 0.10	43.5	50.1	46.0	69.0
R8	340	319	231	42.1[f] ± 0.29	27.9[g] ± 1.1	1.5[g] ± 0.14	71.7	85.7	80.1	71.5
R45	309	58	72	71.0[g]± 0.16	41.1[e] ± 2.3	0.74[d] ± 0.04	25.9	33.0	23.9	67.7
R90	230	28	34	38.7[f] ± 4.4	21.6[d] ± 0.71	2.0[e] ± 0.15	24.6	45.7	34.8	43.5

CH_4: methane; CO_2: carbon dioxide; OM: organic matter; dw: dry weight; MSW: municipal solid waste.

[a]R0: anaerobic reactors with untreated MSW (R0a and R0b are replicate runs); R8, R45, R90: Anaerobic reactors containing MSW that were aerobically pretreated for 8, 45 and 90 days respectively.
[b]NL gas dry kg^{-1} MSW.
[c]Means ± SD based on n = 3 (for OM) and n = 5 (for carbon and nitrogen).
[d–g]All means in the same column that do not share a letter are statistically different at $P < 0.05$ using Tukey's pairwise comparison test; solids reduction was calculated using the ash conservation principle.

days for experiment R0b. During the steady methanogenic phase of R0a and R0b, the CH_4 content remained constantly between 50 and 60% (v/v) until the termination of the experiments, whilst the CO_2 content remained close to 40%. Despite the relatively low biogas generation rates in runs R0a and R0b after 500 days, the steady methanogenic phase was sustained until the termination of these experiments.

However (see Figure 2), the 8-day pretreatment period (reactor R8) led to a quicker establishment of the steady methanogenic phase (i.e. during the first 100 days). In addition, after 300 days

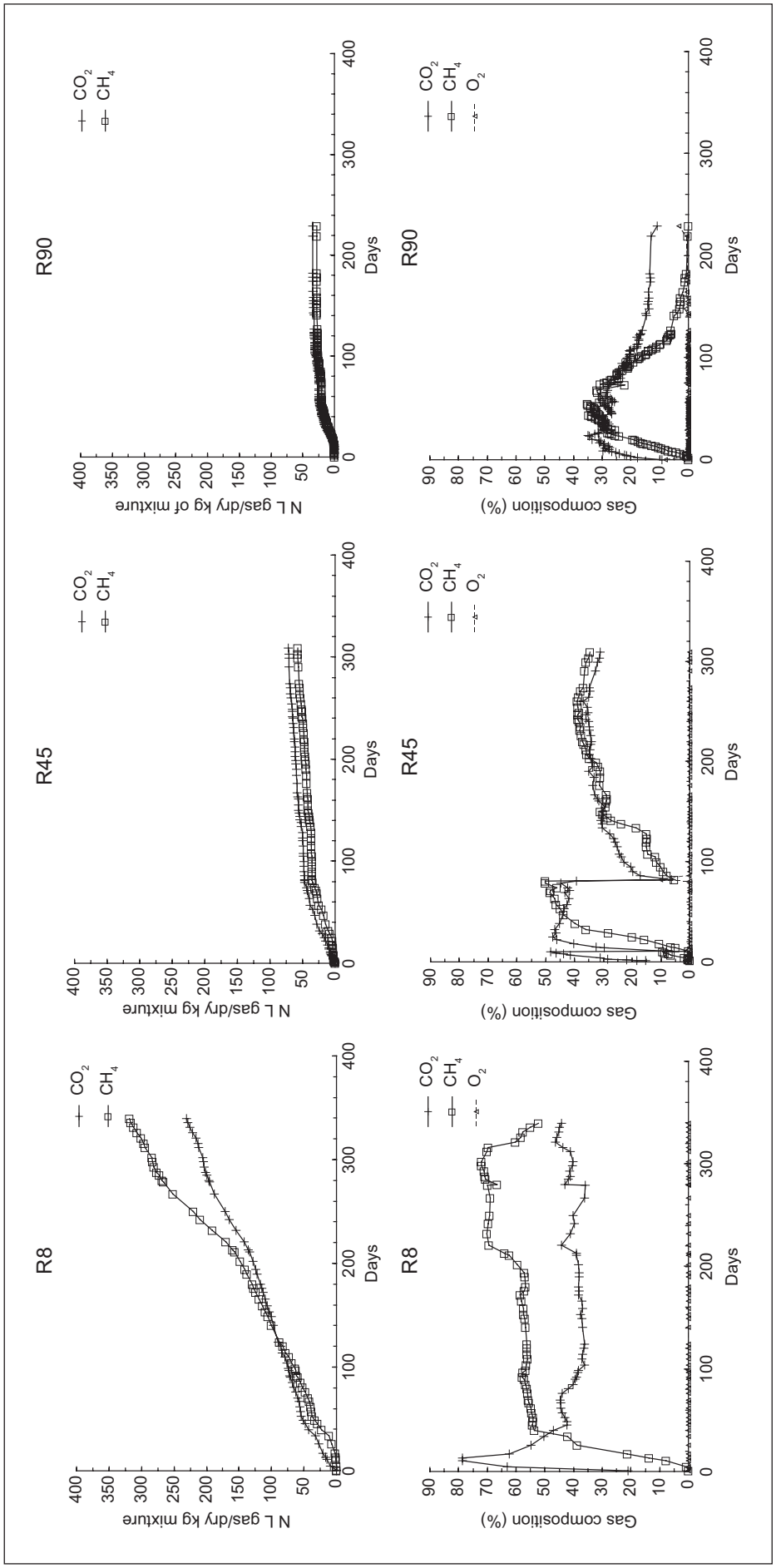


Figure 2. Cumulative methane (CH_4) and carbon dioxide (CO_2) yields (top) and biogas composition (bottom) for anaerobic reactors R8, R45 and R90.

of experimental time, R8 produced greater amounts of biogas compared with both R0 runs over the same time period. This is likely attributed to the fact that the 8-day aerobic pretreatment period removed most of the readily degradable carbon (RDC), which is responsible for the fast acid production and the rapid pH decrease that inhibits methanogenesis (Brummeler and Koster, 1990; Komilis et al., 1999). The 45-day aerobic pretreatment period (see Figure 2) led to an even quicker establishment of the steady methanogenic phase, as the CH_4 content reached values greater than 40% after only 38 days. However, an intermediate opening of R45 for inspection purposes on day 80 temporarily disrupted methanogenesis and led to sharp decreases of the CH_4 and CO_2 contents. Still, CH_4 generation rapidly recovered and reached contents above 30% after day 150. A gradual decrease in the contents of both CH_4 and CO_2 was measured in R45 after the 260th day until the termination of the run.

R90 yielded even smaller amounts of biogas, which reached 62 NL dry kg^{-1} after 230 days. In R90, CH_4 rapidly peaked after 43 days, but never exceeded 35% v/v; it then gradually decreased to values close to 0% until the termination of the experiment. An almost similar profile was also observed for the CO_2. Most of the biogas of R90 was generated during the first 100 days and biogas generation rates decreased significantly after 150 days. The final OM content of the anaerobic residue in R90 was 37% (dw), whilst the dry mass and OM reductions that occurred within that reactor were 25% and 46% respectively (see Table 3). Therefore, it appears that the 45- and 90-day aerobic pretreatment periods led to a 65% and 84% biogas yield reduction compared with the average biogas yield of raw refuse. More specifically, CH_4 yield decreased by 70% and 85% in the R45 and R90 runs, respectively, compared with the average CH_4 yield of R0 (194 NL dry kg^{-1}). CO_2 yields decreased by 60% and 80% in R45 and R90, respectively, compared with that of R0 (178 NL dry kg^{-1}). In addition, the methanogenesis lag times in R45 and R90 were highly shortened (lasted 40 days) compared with that of R0 (160 days). O_2 contents decreased to 0% only a few hours after the initiation of all anaerobic reactors and remained at that level throughout the experiments.

Similar effects of long-term aerobic pretreatment have been found by Binner and Zach (1999), Zach et al. (2000), Jokela et al. (2002) and Corti et al. (2009). For example, Jokela et al. (2002) observed that the CH_4 potential decreased 10 times when an aerobic pretreatment step was first applied (i.e. a decrease from 410 L CH_4 dry kg^{-1} to 41 L CH_4 dry kg^{-1}). However, the results of Jokela et al. (2002) were based on biochemical CH_4 potential tests, which are known to result in higher biogas yields than pilot-scale anaerobic reactors. Other short-term (2-week) aerobic pretreatment studies have also observed a fast establishment of methanogenesis and a reduction of the biogas yield compared with that of raw MSW (Brummeler and Koster, 1990). However, our results show that a very short-term (8-day) aerobic pretreatment of MSW with a high content of putrescibles (which corresponds to a 15% OM reduction) can result not only in the fast establishment of methanogenesis but also in an increase of the biogas yield compared with that of raw MSW over a period of 1.5 years. These findings agree with Adani et al. (2004), who had, however, performed their anaerobic experiments in small vials and used only a 90-day experimental period.

Leachate properties

Figure 3 presents the BOD_5 concentration, the BOD_5/COD ratio, the pH and the $N\text{-}NH_4^+$ concentrations of the leachates versus time for all anaerobic reactors. The average leachate volume collected per reactor at each recirculation event was 0.70 ($R0_a$), 1.3 ($R0_b$), 0.71 (R8), 1.5 (R45) and 0.52 L dry kg^{-1} (R90). According to Figure 3a, the highest BOD_5 concentrations were recorded for the replicate R0a and R0b runs that contained the untreated waste. This is expected, as the RDC of the putrescibles contained in the raw refuse is converted to acids during hydrolysis and acidogenesis. The BOD_5 concentration profiles of the two replicate R0 runs were similar. That is, BOD_5 peaked at 50–100 days as organic acids started to be generated, lowering pH (see Figure 3C). BOD_5 concentrations greater than 20 g L^{-1} were measured during the first 150 days in both R0 runs. BOD_5 then gradually decreased until the 300th day. A slow increasing trend was observed later on in R0 until the termination of the experiment, which could be attributed to the initiation of the breakdown of the moderately degradable organic carbon, which hydrolyses at slower rates than the RDC. However, the BOD_5 contents of all other reactors (R8, R45, R90) remained less than 5 g L^{-1} throughout the experiment. The BOD_5 concentration of the R45, in particular, reached its maximum value on the 10th day (5.1 g L^{-1}) and gradually decreased later on. Similarly, the R90 run observed its maximum BOD_5 content (4.5 g L^{-1}) on the 26th day. According to Figure 3B, there were large fluctuations in the BOD_5/COD ratios for all runs and, particularly, for the R0 runs. For runs R8, R45 and R90, a decreasing trend of the BOD_5/COD ratio is evident, with maximum values at 0.8 (for R90 only) at the beginning of the process and a gradual decrease to less than 0.3 later on. The BOD_5/COD ratio decreasing trends to final values close to 0.1 are typical for aging landfills (McBean et al., 1995, p. 307). The BOD_5/COD ratio of the R0b decreased to values less than 0.2 during the middle of the run, but increased to values greater than 0.7 at the final stages, probably owing to the decomposition of moderately degradable carbon.

The characteristic pH decrease (see Figure 3C) that occurs during anaerobic acidogenesis was recorded during the first 100 days for all runs, except R90. Actually, leachate pH ranged from 4.0 to 5.5 in the R0 runs during the first 80 days, despite the continuous neutralization with Na_2CO_3 over that time. The leachate from the R0a run reached a close-to-neutral pH after 180 days, which was the approximate time of the initiation of the steady methanogenic phase. Similarly, leachate from the R0b reactor reached a neutral pH after 280 days, which was, again, the approximate time of initiation of the steady methanogenic phase for that run. The above agree well with the established landfill degradation theory, which states that methanogenesis in landfills

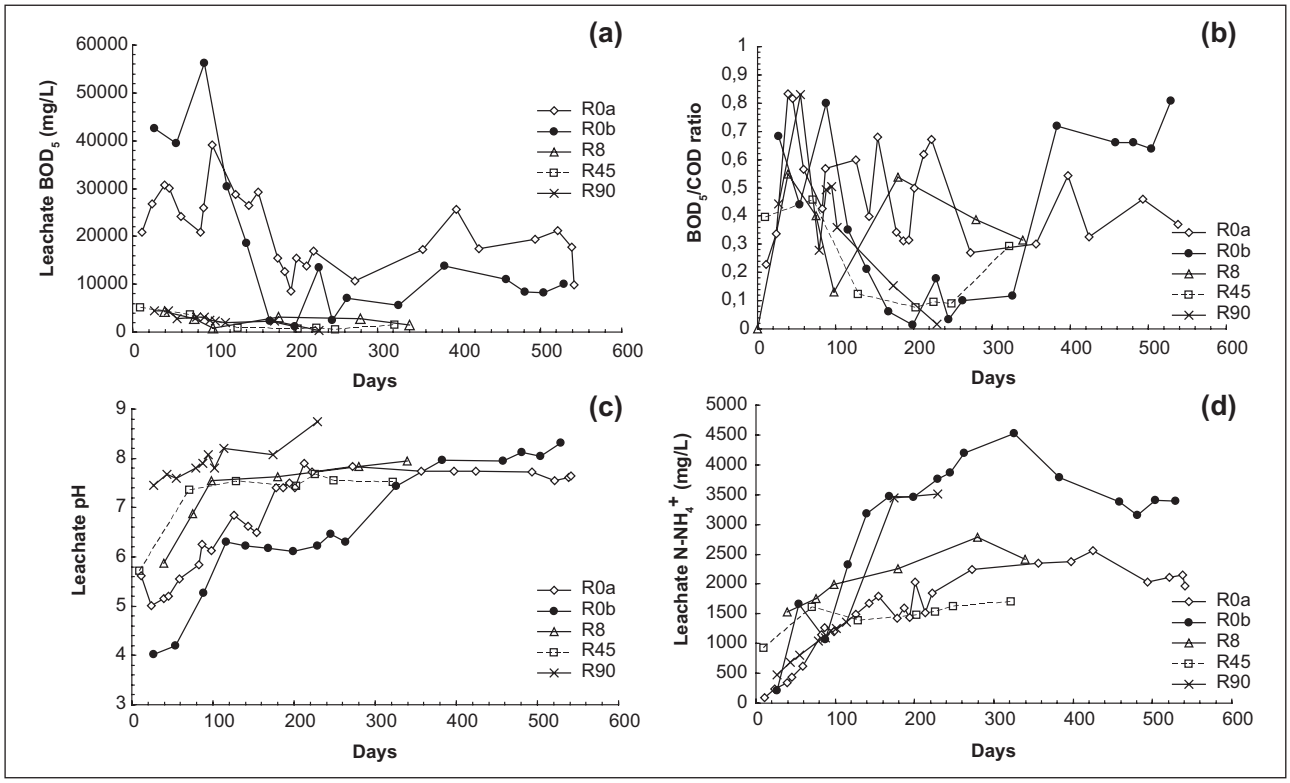


Figure 3. Leachate properties in the five reactors. (A) Five-day biochemical oxygen demand (BOD_5) concentration; (B) BOD_5/chemical oxygen demand (COD) ratio; (C) pH; (D) ammonium ($N\text{-}NH_4^+$) concentration.

starts right after acidogenesis, as acidic pH increases to neutral levels owing to the gradual consumption of acids by methanogens (Farquhar and Rovers, 1997). During the steady methanogenic phase, pH values remained above 7, with a slight increasing trend, for both R0 runs. However, pH started at 5.8 for the R8 and R45 runs, and very rapidly increased and remained above 7 for the rest of the experiment. The leachate pH of R90 was constantly above 7.2, from the start of the experiment, with an increasing trend to up to 8.5 at the end of the run. It therefore appears that a large amount of RDC was removed during the aerobic pretreatment stage, which prevented excessive acid formation during the subsequent anaerobic stage.

The $N\text{-}NH_4^+$ concentrations for all runs observed an increasing trend with time until a plateau was always reached. This ammonium accumulation has also been observed by other researchers (Mahar et al., 2009), and is explained by the low biomass yield during anaerobiosis and the lack of nitrification. Although some ammonia (NH_3) gaseous emissions may occur (though NH_3 was not measured in this work), the lack of the above processes inevitably led to NH_4^+ accumulation. As indicated in Figure 3D, the NH_4^+ concentration in both R0 runs observed an increasing trend from 0.05 g L^{-1} up to 4.5 g L^{-1} (for R0b). R0b resulted in leachate that had almost twice as high as NH_4^+ concentration compared with the R0a, although the trends were similar for both runs. The NH_4^+ of the leachates from the R8 and R45 runs also observed an increasing trend to concentrations of up to 2.5 g L^{-1} (R8). Strangely, the NH_4^+ content in R90 almost doubled during the last two sampling events to values up to 3.5 g L^{-1}. This is probably attributed to the low amounts of leachate generated at the end of the R90 run, which led to an inevitable concentration of the NH_4^+ ions. The NH_4^+ concentration in the leachate of R45 reached a plateau at 1.5 g L^{-1} after 100 days.

Some similarities of the 8-day pretreatment stage with the two-stage anaerobic digestion process do exist. In the two-step anaerobic digestion, hydrolysis is performed in a separate reactor and the hydrolyzed material is then supplied under control to the methanogenic reactor so as not to inhibit methanogenesis from a rapid pH decrease (Merlino et al., 2013). In our case, the 8-day aerobic pretreatment reactor hydrolyzed some of the readily degradable organics to sugars and acids, which were eventually decomposed to CO_2. Thus, less readily degradable carbon was loaded to the subsequent anaerobic reactors and methanogenesis inhibition was prevented.

Decrease of solids during the aerobic and anaerobic processes

Figure 4A shows that the highest OM content decrease occurred during the 90-day aerobic pretreatment step (i.e. OM decreased from 86% dw to 53% dw), which corresponds to an 80% decrease (see Table 2). The OM content of the anaerobic end product of R90 was close to 38% dw, the lowest value among all OM contents of the anaerobic residues, then followed the end product of R8 with an OM content of 42.1% (dw). Figure 4B shows the relationship between the biogas yields of all five anaerobic reactors with the OM decrease achieved during the ABT phase.

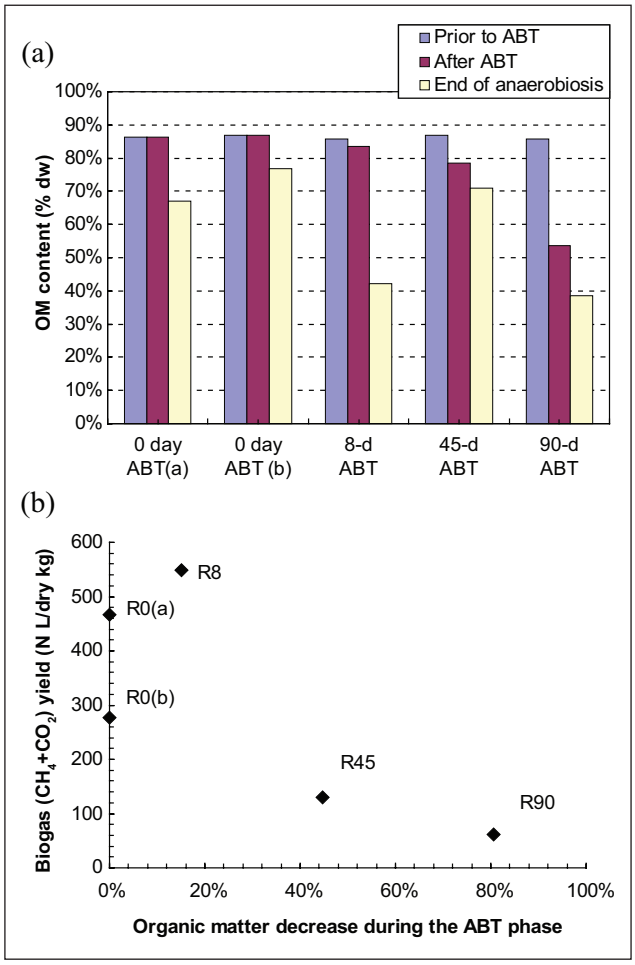


Figure 4. (A) Organic matter (OM) content at the start and end of the aerobic pretreatment (ABT) and anaerobic phases; (B) biogas yield versus the organic matter reductions achieved during the aerobic pretreatment stage. CH_4: methane; CO_2: carbon dioxide.

According to Figure 4B, the short-term (8-day) aerobic pretreatment step (15% OM decrease) led to an enhancement of biogas generation compared with raw refuse, which partly agrees with Adani et al. (2004), who had also used short-term aerobic pretreatment periods. However, higher OM reduction (above 40%) led to decreased biogas yields, as has been clearly shown by several researchers (Binner and Zach, 1999; Carlsson et al., 2012; De Gioannis et al., 2009; Mahar et al., 2009; Zach et al., 2000). Therefore, an at least 40% OM decrease has to be achieved during an ABT phase to generate waste for disposition to a zero emission landfill. However, if the goal is to enhance biogas generation, then the OM decrease of MSW with high amounts of putrescibles should not exceed 15%. Although some carbon is inevitably lost during the ABT phase, the benefits from the enhanced biogas yields during the first 1–1.5 years would offset the carbon lost.

Conclusions

- A very short-term (8 day) aerobic pretreatment period (that corresponded to a 15% OM loss) accelerated the start-up of methanogenesis, resulted in non-acidic leachates and led to higher biogas yields compared with that of raw MSW.
- The 45% and 81% OM reduction (that corresponded to 45- and 90-day aerobic pretreatment periods respectively) reduced the biogas yield of untreated waste by 65% and 84% respectively. In addition, the lag time before the onset of methanogenesis was 4–5 times shorter in the pretreated waste compared with that of untreated waste.
- The leachate BOD_5 concentrations of the pretreated waste were all below 5 g L^{-1}, as opposed to that of untreated waste, which reached maximum concentrations up to 55 g L^{-1}. Leachate NH_4^+ concentrations showed an increasing trend for all runs until they reached a plateau that ranged from 1.5 g L^{-1} to 3.5 g L^{-1}.
- As the aerobic pretreatment period increased, the pH of the landfill leachates became less acidic.

Acknowledgements

We wish to acknowledge the following individuals for their support during this project: K. Blouchou, G. Kosmidis, V. Lazaridis, V. Papanastassiou, S. Tsipissi, I. Papaspyros and Professor E. Voudrias. Dimitrios Komilis dedicates this work to the memory of his former PhD advisor, the late Bob Ham.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

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